

Burden and Outcomes of Dengue-Related Hospitalizations in the United States: A Proposed Secondary Analysis of National Inpatient Data

Abstract

Dengue is a rapidly expanding vector-borne viral disease and a growing global public health concern, with an estimated 390 million infections annually, of which approximately 96 million are clinically apparent (Bhatt *et al.*, 2013). Although many infections are self-limited, approximately 5–10% progress to severe dengue, characterized by complications such as hemorrhage, shock, and organ dysfunction (WHO, 2024). Increasing international travel, climate change, and expansion of *Aedes* mosquito vectors have heightened the risk of dengue emergence in the United States.

This study aims to evaluate the demographic characteristics, comorbidity profiles, clinical outcomes, and healthcare utilization associated with dengue-related hospitalizations in the United States using nationally representative inpatient data. A retrospective observational design will be employed using the Healthcare Cost and Utilization Project (HCUP) Nationwide Inpatient Sample. Outcomes of interest include ICU admission, length of stay, and in-hospital mortality.

Findings from this study are expected to provide nationally relevant estimates of dengue-related inpatient burden and inform public health preparedness, clinical management, and risk stratification strategies.

1. Introduction

Dengue fever is a mosquito-borne viral disease caused by the dengue virus (DENV) and transmitted primarily by *Aedes* mosquitoes. It represents a major global public health challenge, with approximately 390 million infections annually (Bhatt *et al.*, 2013). While most infections are mild, about 5–10% progress to severe dengue, leading to significant morbidity and mortality (WHO, 2024).

Dengue is increasingly recognized as an emerging concern in the United States due to factors such as international travel, climate change, and the geographic expansion of mosquito vectors (CDC, 2024). Reports of locally acquired cases in states such as Florida and Texas highlight the potential for domestic transmission (Radke *et al.*, 2012). Despite this, the burden and outcomes of dengue-related hospitalizations in the United States remain poorly characterized, particularly among high-risk populations such as older adults and individuals with comorbidities.

This report aims to examine the burden and clinical outcomes of dengue-related hospitalizations in the United States, with a focus on demographic patterns, healthcare utilization, and implications for public health preparedness.

2. Impact on Population and Public Health

Dengue imposes a significant burden on global health systems due to its high incidence, potential for severe complications, and economic impact. The disease contributes substantially to morbidity and healthcare utilization, particularly in endemic regions (Stanaway *et al.*, 2016). In addition to clinical consequences, dengue outbreaks result in considerable economic costs due to hospitalization, vector control, and productivity loss (Shepard *et al.*, 2016).

In the United States, dengue has historically been limited to imported cases; however, increasing evidence suggests a shift toward local transmission. Factors such as climate change, urbanization, and increased travel have contributed to the spread of *Aedes* mosquitoes and heightened transmission risk (Ryan *et al.*, 2019). Regions such as Florida, Texas, and Puerto Rico are particularly vulnerable due to favorable environmental conditions (CDC, 2024).

Dengue-related hospitalizations place a growing burden on healthcare systems through increased admissions, longer hospital stays, and the need for intensive care in severe cases. Vulnerable populations, including older adults and individuals with chronic diseases, are at increased risk for adverse outcomes (Guzman and Harris, 2015). From a public health informatics perspective, effective surveillance, data integration, and predictive modeling are essential for monitoring disease trends and informing response strategies.

3. Challenges Associated with Dengue Research

Several challenges limit the accurate assessment of dengue burden and outcomes in the United States. First, underreporting and misdiagnosis are common due to symptom overlap with other febrile illnesses and limited clinical suspicion in non-endemic regions (Roy and Bhattacharjee, 2021).

Second, there is limited availability of U.S.-specific data on dengue hospitalizations, with most research focused on endemic countries. This creates a significant gap in understanding the domestic impact of the disease.

Third, reliance on administrative data introduces challenges such as coding inaccuracies and misclassification when using ICD-10 diagnostic codes. These limitations may affect the validity of epidemiological estimates.

Additionally, environmental unpredictability related to climate change complicates efforts to predict dengue transmission patterns. The expansion of mosquito vectors into new regions further increases uncertainty (Messina *et al.*, 2019).

Finally, gaps in surveillance systems and data integration hinder real-time monitoring and response, limiting the effectiveness of public health interventions.

4. Related Work (Literature Review)

Extensive research has been conducted on the global epidemiology and burden of dengue. Bhatt *et al.* (2013) provided one of the most comprehensive estimates of global dengue incidence, highlighting significant underreporting in surveillance systems. Similarly, Stanaway *et al.* (2016) quantified the global burden using disability-adjusted life years (DALYs), demonstrating a substantial increase in disease impact over time.

Studies on dengue pathogenesis, such as Roy and Bhattacharjee (2021), have explored the biological mechanisms underlying disease progression and severity. These studies emphasize the complexity of dengue infection and the factors contributing to severe outcomes.

Research in non-endemic regions, including the United States, has primarily focused on outbreak investigations and surveillance reports. For example, Radke *et al.* (2012) documented local transmission in Florida, highlighting the potential for dengue emergence in the U.S.

However, there is a lack of comprehensive studies examining hospitalization outcomes at a national level in the United States. Existing literature does not adequately address healthcare utilization, comorbidity profiles, and inpatient outcomes, underscoring the need for further research in this area.

5. Methodological Framework

This study employs a retrospective observational design using secondary data from the Healthcare Cost and Utilization Project (HCUP) Nationwide Inpatient Sample (AHRQ, 2024). The dataset provides nationally representative information on hospitalizations across the United States.

The study population includes adult patients (≥ 18 years) hospitalized with a primary or secondary diagnosis of dengue, identified using ICD-10-CM codes. Variables of interest include demographic characteristics, comorbidities, and hospital-level factors.

Primary outcomes include ICU admission, length of hospital stay, and in-hospital mortality. Secondary outcomes include severe dengue complications such as hemorrhage, shock, and organ dysfunction.

Statistical analysis will include descriptive statistics and multivariable regression models to assess associations between risk factors and outcomes. Sampling weights will be applied to generate nationally representative estimates. Ethical considerations are addressed through the use of de-identified secondary data, qualifying the study for IRB exemption.

6. Opportunities for Future Research

Future research should focus on enhancing dengue surveillance systems through improved data integration and real-time monitoring. The application of machine learning and predictive analytics offers opportunities for early outbreak detection and risk assessment.

Climate-based modeling can help predict the geographic expansion of dengue and inform targeted interventions (Ryan *et al.*, 2019). Additionally, further U.S.-focused studies are needed to evaluate hospitalization patterns, comorbidity effects, and regional variations.

Research on vaccine effectiveness and therapeutic interventions remains critical, particularly in the context of emerging transmission in non-endemic regions. Finally, addressing health disparities and improving healthcare access will be essential for reducing the burden of dengue among vulnerable populations.

7. News, Data and Current Events

Recent global trends indicate a significant increase in dengue cases, with multiple regions experiencing record outbreaks. Climate change, urbanization, and globalization have contributed to the expansion of mosquito habitats and increased transmission (WHO, 2024).

In the United States, dengue activity has been reported in states such as Florida and Texas, with both imported and locally acquired cases (CDC, 2024). These developments highlight the growing relevance of dengue as a domestic public health issue.

Public health responses include enhanced surveillance, vector control measures, and public education campaigns aimed at reducing transmission risk. These efforts underscore the importance of data-driven strategies and preparedness in addressing emerging infectious diseases.

8. Overall Assessment and Conclusion

Dengue fever represents an emerging and under-recognized public health threat in the United States. While global research has extensively documented its burden, there remains a significant gap in understanding its impact on the U.S. healthcare system.

This study provides a framework for evaluating dengue-related hospitalizations using nationally representative data, offering valuable insights into clinical outcomes, healthcare utilization, and risk factors. The findings highlight the need for improved surveillance, enhanced healthcare preparedness, and targeted public health interventions.

As environmental and societal factors continue to evolve, dengue is likely to pose an increasing challenge in the United States. A multidisciplinary approach integrating epidemiology,

informatics, and policy will be essential for mitigating its impact and protecting population health.

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